

TRISTAN ALDERSON

Work Authorization: Dual US & Canadian Citizen

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EDUCATION

Bachelor of Applied Science in Electrical and Computer Engineering
Queen's University • *Dean's Scholar*

Kingston, ON, Canada
05/2026

HARDWARE & EMBEDDED FOCUS

Hardware/embedded engineer focused on PCB design and bring-up, **STM32/ESP32** firmware, **CAN** avionics, power distribution, GPS/RF integration, and Linux/FPGA hardware interfaces; comfortable debugging across schematic, power rails, boot behavior, firmware, telemetry, and physical system behavior.

WORK EXPERIENCE

Avionics Lead

Queen's Rocket Engineering Team (QRET)

05/2025 – Present
Kingston, ON, Canada

- Lead a 10-person avionics team building hybrid-rocket flight electronics across **STM32/ESP32** control nodes, **CAN** networking, power distribution, telemetry, GPS recovery, and launch-ops ground-station hardware.
- Designed and brought up **STM32** and **ESP32** control/sensor boards for solenoid actuation, pressure-transducer reads, switched sensor power, CAN reporting, and hardware fail-safes; managed schematics, PCB layout, JLCPCB assembly, bench validation, and manual rework.
- Engineered a **24V avionics power module** with ORed ground/battery inputs, **3.3V/5V buck rails**, PMOSFET reverse-polarity protection, and zener Vgs clamping; diagnosed an overheating zener shunt during current-limited first power-on and reworked the board.
- Architected **AimNetwork**, a fixed-format **CAN** protocol with 64-bit payloads, strict node addressing, heartbeat checks, time synchronization, and WiFi/LoRa telemetry forwarding across rocket avionics nodes.

Embedded Software Engineer Intern

Evertz Microsystems

05/2024 – 08/2025
Burlington, ON, Canada

- Built distributed **C++ Linux** control services for a 1RU embedded hardware platform, coordinating device status, configuration, telemetry, and control across **six ARM Linux nodes** using gRPC, REST APIs, and Redis-backed state.
- Developed Linux application and monitor software for ARM front-panel devices, including **SFP/QSFP** state, **NTP/PTP** synchronization, system status, remote configuration, and event logging.
- Debugged 1G/10G/100G **SFP/QSFP** paths, multicast forwarding, TLS/OpenSSL failures, and Marvell switch behavior using vendor tooling, **GDB**, logs, board traces, and custom ACL rules.

AI Engineer & Avionics Developer

Queen's Rocket Engineering Team (QRET)

10/2023 – 08/2024
Kingston, ON, Canada

- Designed and flew a custom **SAM-M10Q GPS** daughter board with I2C breakout, physical separation from comms electronics, and enlarged ground plane to improve reception and reduce RF interference.
- Reworked a handheld **ground-station receiver** for live telemetry decode and GPS recovery tracking, including a simplified 5V power path and 330mF supercapacitor backup to preserve GPS state through brief power loss.
- Secured **2nd Place at Launch Canada** by implementing YOLOv8 rocket tracking with dual-axis PID control and Matlab simulation; owned 18650 battery, altimeter, and GPS configuration during launch operations.

Programmer and Assembler

Radius Security

05/2023 – 09/2023
Richmond, BC, Canada

- Manufactured remote-guarding surveillance units; earned internal recognition for wiring quality and contributed to **Lean** process improvements that increased manufacturing space and overall efficiency by **20%**.

PROJECTS

LiteLM: FPGA-backed LLM Inference on ARM Linux

Queen's University

08/2025 – 03/2026
Kingston, ON, Canada

- Earned **3rd Place for Top Computer Engineering Project** at the Queen's ECE Showcase for GPT-2 inference on an **AMD Xilinx Kria KV260** using FPGA fabric for matrix operations and ARM Cortex-A53 Linux firmware for precision-sensitive operations.
- Implemented hardware/software interfaces using **AXI4 DMA**, DDR memory maps, device trees, U-Boot boot files, hardware overlay loading, and register access between processing system and programmable logic.
- Debugged DMA-backed accelerator buffers and Linux-side transfer stability across CMA sizing, udmabuf allocation, boot-time service behavior, and device-tree configuration.

SKILLS

Hardware & EDA: KiCad, LTspice, Vivado, Vitis HLS, JLCPCB assembly, schematic capture, PCB layout, manual rework, oscilloscopes, logic analyzers, current-limited supplies

Embedded platforms: STM32, ESP32, Zynq UltraScale+, Kria KV260, ARM Cortex-A53, ARM Linux, x86 COM Express
Power, RF & interfaces: DC-DC buck rails, PMOSFET protection, zener clamps, 18650 packs, GPS, RF layout, ground planes, CAN, I2C, SPI, UART, LoRa, WiFi

Languages & firmware: C, C++, Python, Bash, Matlab, Verilog/SystemVerilog, STM32 HAL, ESP-IDF, fixed queues, heartbeat logic

Linux & debug: Embedded Linux, Buildroot, U-Boot, device trees, AXI4, AXI DMA, DDR memory maps, GDB, Git, logs, board bring-up, fault isolation